

Lezione del 7 maggio 2026

Thursday, May 7, 2026 12:56 PM

$$Y \left\{ \begin{array}{l} \{-3, \dots, 5\} \\ \{\frac{1}{9}, \dots, \frac{1}{9}\} \end{array} \right\}$$

$$Z = |X| \quad \{-2, -1, 0, 1, 2, 3, 4, 5, 6\}$$

$$\begin{array}{l} \{2, 1, 0, 1, 2, 3, 4, 5, 6\} \\ \left\{ \frac{1}{9}, \frac{1}{9}, \frac{1}{9}, \frac{1}{9}, \frac{1}{9}, \frac{1}{9}, \frac{1}{9}, \frac{1}{9}, \frac{1}{9} \right\} \end{array}$$

$$\{0, 1, 2, 3, 4, 5, 6\}$$

$$\left\{ \frac{1}{9}, \frac{2}{9}, \frac{2}{9}, \frac{1}{9}, \frac{1}{9}, \frac{1}{9}, \frac{1}{9} \right\}$$

$$E[Z] = 1 + \frac{2}{9} + 2 \times \frac{2}{9} + 3 \times \frac{1}{9} + \dots + 6 \times \frac{1}{9}$$

$$E[|X|] = |-2| \times \frac{1}{9} + |-1| \times \frac{1}{9} + |0| \times \frac{1}{9} + |1| \times \frac{1}{9} + |2| \times \frac{1}{9} +$$

$$+ |3| \times \frac{1}{9} + |4| \times \frac{1}{9} + |5| \times \frac{1}{9} + |6| \times \frac{1}{9}$$

$$E[|XZ|] = E[|X||X|] = (-2) \times \frac{1}{9} + (-1) \times \frac{1}{9}$$

$$+ 0 \times \frac{1}{9} + 1 \times \frac{1}{9} + 2 \times \frac{1}{9} + 3 \times \frac{1}{9}$$

$$+ 4 \times \frac{1}{9} + 5 \times \frac{1}{9} + 6 \times \frac{1}{9}$$

$X \backslash Y$	y_1	y_2	
x_1	a	b	$a+b$
x_2	c	d	$c+d$
	$a+c$	$b+d$	

$$a, b, c, d > 0$$

$$a+b+c+d = 1$$

$$H(X, Y) =$$

$$(-a \log_2 \frac{1}{a} + b \log_2 \frac{1}{b} + c \log_2 \frac{1}{c} + d \log_2 \frac{1}{d})$$

$$H(X) = (a+b) \log_2 \frac{1}{a+b} + (c+d) \log_2 \frac{1}{c+d}$$

$$P[X=x_1] = P[X=x_1, Y=y_1] P[Y=y_1] + P[X=x_1, Y=y_2] P[Y=y_2]$$

$$= P[X=x_1, Y=y_1] + P[X=x_1, Y=y_2]$$

$$= a + b$$

$$P[X=x_2] = P[X=x_2, Y=y_1] + P[X=x_2, Y=y_2] = c + d$$

$$I(X, Y) = \sum_{x,y} p(x,y) \log_2 \frac{p(x,y)}{p_x(x) p_y(y)}$$

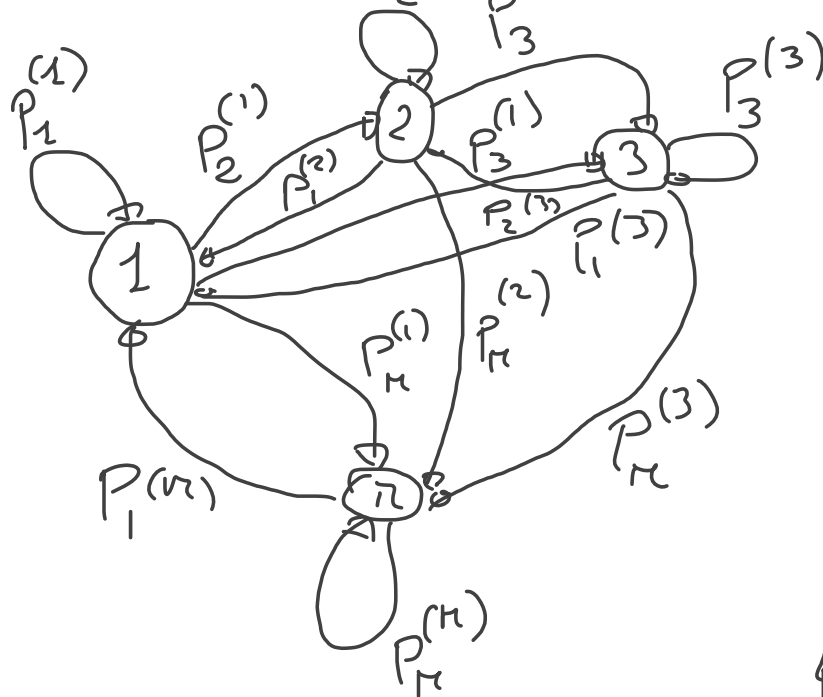
$$= a \log_2 \frac{a}{(a+b)(a+c)}$$

$X \backslash Y$	y_1	y_2
x_1	$(a+b)(a+c)$	$(a+b)(b+d)$
x_2	$(c+d)(a+c)$	$(c+d)(b+d)$

$$+ b \log_2 \frac{b}{(a+b)(b+d)} + c \log_2 \frac{c}{(c+d)(a+c)} + d \log_2 \frac{d}{(c+d)(a+d)}$$

$$P[X(n)=i | X(n-1)=j] = P_i^{(j)}$$

$$X(n-1)=1 \quad P[X(n)=i] = P_i^{(1)}$$



$$P[X(n)=i | X(n-1)=2] = P_i^{(2)}$$

$$P[X(n)=i | X(n-1)=3] = P_i^{(3)}$$

$$\begin{bmatrix} P_2^{(1)} & P_2^{(1)} & \dots & P_n^{(1)} \\ P_1^{(2)} & P_2^{(2)} & \dots & P_n^{(2)} \\ \vdots & \vdots & \ddots & \vdots \\ P_1^{(n)} & P_2^{(n)} & \dots & P_n^{(n)} \end{bmatrix}$$

$$[P_{11} \ P_{12} \ \dots \ P_{1n}]$$

$$\begin{bmatrix} P_{11} & P_{12} & \dots & P_{1n} \\ P_{21} & P_{22} & & P_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ P_{n1} & P_{n2} & & P_{nn} \end{bmatrix}$$

$$P_{ki} = P_i^{(k)}$$

$$f_{X(n)}(i) = P[X(n)=i] = \sum_{j=1}^M P[X(n)=i | X(n-1)=j] P[X(n-1)=j]$$

$$\underbrace{P_i[n]}_{f(n)} = \sum_{j=1}^M \underbrace{P_{ij}^{(n)}}_{P^T(n)} \underbrace{P_j[n-1]}_{f(n-1)}$$

$$\underbrace{\begin{bmatrix} P_i[n] \end{bmatrix}}_{f(n)} = \underbrace{\begin{bmatrix} P_{i1}^{(n)} & P_{i2}^{(n)} & \dots & P_{in}^{(n)} \end{bmatrix}}_{P^T(n)} \underbrace{\begin{bmatrix} P_1[n-1] \\ P_2[n-1] \\ \vdots \\ P_n[n-1] \end{bmatrix}}_{f(n-1)}$$

use for eqn i

$$\begin{bmatrix} P_{i1}^{(n)} & P_{i2}^{(n)} & \dots & P_{in}^{(n)} \end{bmatrix} = \begin{bmatrix} P_i^{(1)}(n) \\ \vdots \\ P_i^{(n)}(n) \end{bmatrix}$$

$$f(n) = \underbrace{P^T(n)}_{\varphi} f(n-1)$$

1, 1, 0, 1

1. CATENA ONOGENEA: $\underline{f}(n) = \underline{f}$

$$\underline{f}(n) = \underline{P}^T \underline{f}(n-1) = \underline{P}^T \left[\underline{P}^T \underline{f}(n-2) \right] = (\underline{P}^2)^T \underline{f}(n-2)$$

$$= (\underline{P}^2)^T \left[\underline{P}^T \underline{f}(n-3) \right] = (\underline{P}^3)^T \underline{f}(n-3) = (\underline{P}^i)^T \underline{f}(n-i) \quad \forall i$$

$$i_n, i_{n-1} \in \{1, \dots, M\}$$

$$\underline{X}_n = \begin{bmatrix} \underline{X}(n-1) \\ \underline{X}(n) \end{bmatrix} \quad \begin{matrix} \uparrow & \uparrow \\ n-1 & n \end{matrix} \quad \underline{y}_n = \begin{bmatrix} i_{n-1} \\ i_n \end{bmatrix}$$

$$P[\underline{X}_n = \underline{y}_n] = P[X(n) = i_n, X(n-1) = i_{n-1}]$$

$$= \underbrace{P[X(n-1) = i_{n-1}]}_{\underline{f}_{i_{n-1}}(n)} P[X(n) = i_n | X(n-1) = i_{n-1}]$$

$$\underline{f}_{i_{n-1}}(n-1) \quad \begin{matrix} \uparrow & \uparrow & \uparrow \\ n-2 & n-1 & n \end{matrix} \quad P_{i_{n-1}, i_n}$$

$$X(n) \quad X(n-1) \quad X(n-2)$$

$$i_n, i_{n-1}, i_{n-2} \in \{1, \dots, r\}$$

$$\underline{X}(n) = \begin{bmatrix} X(n-2) \\ X(n-1) \\ X(n) \end{bmatrix} \quad \underline{x}_n = \begin{bmatrix} i_{n-2} \\ i_{n-1} \\ i_n \end{bmatrix}$$

$$P[\underline{X}(n) = \underline{x}_n] = P[X(n) = i_n, X(n-1) = i_{n-1}, X(n-2) = i_{n-2}]$$

$$= P[X(n-2) = i_{n-2}] P[X(n-1) = i_{n-1} | X(n-2) = i_{n-2}]$$

$$P[X(n) = i_n | X(n-1) = i_{n-1}, X(n-2) = i_{n-2}]$$

$$= P[X(n-2) = i_{n-2}] P[X(n-1) = i_{n-1} | X(n-2) = i_{n-2}] \quad \ell=2 \rightarrow \begin{bmatrix} i_{n-1} \\ i_n \end{bmatrix}$$

$$P[X(n) = i_n | X(n-1) = i_{n-1}] \quad \ell=3 \rightarrow \begin{bmatrix} i_{n-2} \\ i_{n-1} \\ i_n \end{bmatrix}$$

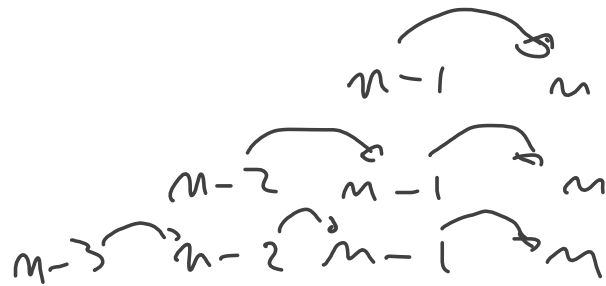
$$P[\underline{X}_{n-\ell+1:n} = \underline{x}_{n-\ell+1:n}]$$

$$\underline{x}_{n-\ell+1:n} = \begin{bmatrix} i_{n-\ell+1} \\ \vdots \\ i_{n-2} \\ i_{n-1} \\ i_n \end{bmatrix}$$

$$= P[X(n-\ell+1) = i_{n-\ell+1}] \prod$$

SACTI 5020

l-1



$$l=2$$

$$l=3$$

$$l=4$$

$$\underline{X}_n = \begin{bmatrix} X(n-3) \\ X(n-2) \\ X(n-1) \\ X(n) \end{bmatrix} \quad \underline{I}_n = \begin{bmatrix} i_{n-3} \\ i_{n-2} \\ i_{n-1} \\ i_n \end{bmatrix}$$

$$P[\underline{X}_n = \underline{x}_n] = P[X(n-3) = i_{n-3}, X(n-2) = i_{n-2}, X(n-1) = i_{n-1}, X(n) = i_n]$$

$$= P[X(n-3) = i_{n-3}] P[X(n-2) = i_{n-2} | X(n-3) = i_{n-3}]$$

9
l-1

$$P[X(n-1) = i_{n-1} | X(n-2) = i_{n-2}, X(n-3) = i_{n-3}]$$

$$P[X(n) = i_n | X(n-1) = i_{n-1}, X(n-2) = i_{n-2}, X(n-3) = i_{n-3}]$$

$$P[\underline{X}_n = \underline{x}_n] = P[X(n-3) = i_{n-3}] \prod_{k=0}^2 P[X(n-k) = i_{n-k} | X(n-k-1) = i_{n-k-1}]$$

$$p(n+1) = p^T p(n)$$

$$\underline{P}^{-1} \underline{P} = \underline{I} \quad \underline{P} \underline{P}^{-1} = \underline{I}$$

$$\underline{f}(n_0+2) = \underline{P}^T \underline{f}(n_0+1) = (\underline{P}^T)^T \underline{f}(n_0) \quad (\underline{AB})^T = \underline{B}^T \underline{A}^T$$

$$\underline{f}(n_0+n) = (\underline{P}^T)^T \underline{f}(n_0) \quad (\underline{P}^T)^2 = (\underline{P}^T)(\underline{P}^T) = (\underline{P}^2)^T$$

$$(\underline{P}^T)^n = (\underline{P}^T)^{n-1} \underline{P}^T = (\underline{P}^{n-1})^T \underline{P}^T = [\underline{P} \cdot \underline{P}^{n-1}]^T = (\underline{P}^n)^T$$

$$\underline{v} = \underline{P}^T \underline{u}$$

$$\underline{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$$

$$v_1, v_2, \dots, v_n \geq 0$$

$$\sum_{i=1}^n v_i = 1$$

$$\underline{w} = \underline{P}^T \underline{x}$$

$$\underline{w} = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix}$$

$$w_1, w_2, \dots, w_n \geq 0$$

$$\underline{w} \neq \underline{0}$$

$$A = \sum_{i=1}^n w_i > 0$$

$$\underline{v} = \underline{P}^T \left[\frac{1}{A} \underline{x} \right] = \frac{1}{A} \underline{P}^T \underline{x}$$

$$= \frac{1}{A} \underline{w} = \underline{v}$$

$$\underline{v} = \frac{1}{A} \underline{w} =$$

$$\begin{bmatrix} \frac{w_1}{A} \\ \frac{w_2}{A} \\ \vdots \\ \frac{w_n}{A} \end{bmatrix}$$

$$\rightarrow \frac{w_1}{A} + \frac{w_2}{A} + \dots + \frac{w_n}{A}$$

$$= \frac{w_1 + w_2 + \dots + w_n}{A} = 1$$

$$(A_1 A_2) A_3 = A_1 (A_2 A_3)$$

$$A v_i = \lambda_i v_i$$

$$\lambda \underline{I} - \underline{P}^T =$$

$$\begin{bmatrix} 0 - p_1^{(i)} \\ 0 - p_2^{(i)} \\ \vdots \end{bmatrix}$$

. = -

-

[

$$\begin{matrix} \lambda - P_i^{(c)} \\ 0 & - \\ 0 & -P_n^{(c)} \end{matrix}$$

$$\lambda - 1$$